

FINAL INTEGRATION REPORT - Team 05

- CA4 Critical Design Review (CDR) MC3113 Mechatronic Systems Design
- Date: 31 May 2026
- Author: C.H Diyunuge (Architecture & Final Report Lead)
- CA1 Standard: ISO 9241-210 (Human-Centred Design)

Team Members:

Member ID	Name	Role
43	S.M.N Kulasinghe	Requirements & Verification Lead
46	C.H Diyunuge	Architecture & Final Report Lead
53	K.V Kamalasuriya	Safety & Risk Analysis Lead
180	Pasindu Adikari	Controller & HMI Lead

1. INTRODUCTION

1.1 Project Overview

This project developed a digital twin for an autonomous line-following robot, following the VDI 2206 V-model through three sequential design review gates:

- CA2 (SDR): System Design Review - Requirements, architecture, initial controller
- CA3 (PDR): Preliminary Design Review - S1 verification, S2/S3 preliminary analysis
- CA4 (CDR): Critical Design Review - Full verification, HMI, safety evidence

The system integrates PID control, sensor fusion, obstacle avoidance, and fault handling, validated across three scenarios:

- S1 (Nominal): Line following on straight and curved track
- S2 (Obstacle): Static obstacle avoidance at $x = 30\text{m}$
- S3 (Fault): Sensor dropout with speed cap and recovery

1.2 Team Roles

Member	Role	Responsibilities
43 (Kulasinghe)	Requirements & Verification Lead	RTM, verification report, metrics.json, plots
46 (Diyunuge)	Architecture & Final Report Lead	Final integration report, README.md
53 (Kamalasuriya)	Safety & Risk Analysis Lead	Safety evidence, fault procedure, risk log
180 (Adikari)	Controller & HMI Lead	Controller tuning, HMI, final logs

1.3 CA1 Standard Integration

Our team studied ISO 9241-210 (Human-Centred Design for Interactive Systems) in CA1. This standard emphasizes:

- User needs identification (Clause 5.2)
- User safety in error conditions (Clause 7.3)
- Usability evaluation (Clause 8.3)

These principles shaped our requirements, HMI design, and fault handling approach throughout CA2-CA4.

2. REQUIREMENTS → DESIGN JOURNEY

2.1 How ISO 9241-210 Influenced Requirements

Clause	Requirement	Influence
5.2 (User needs)	R-HMI-01 (10Hz telemetry)	Users need real-time feedback
7.3 (Error safety)	R-SAFE-01 (Speed cap)	Safe state during sensor failure
8.3 (Usability)	R-HMI-02 (Scenario selector)	Intuitive scenario selection

The FMEA mindset of "what can go wrong?" shaped our S3 fault scenario design.

2.2 Controller Design Decision

We chose PID over state-space control based on Pugh matrix analysis (CA2):

Criterion	Weight	PID	State-Space
Performance (IAE)	40%	8	9
Tuning Ease	25%	9	6
Computational Cost	20%	9	7
Robustness	15%	7	8
Weighted Score	-	8.35	7.65

Decision: PID selected due to superior tuning ease and adequate performance. **Validation:** PID achieved $IAE = 0.021 \text{ m}\cdot\text{s}$ (< 0.022 target) with 3 weeks of iterative tuning.

2.3 Key DfX Trade-offs

Design for X (DfX)	Priority	Decision
Performance	40% weight	PID with $K_p=2.3$, $K_d=0.13$
Energy Efficiency	20% weight	$v_{\text{base}} = 0.52 \text{ m/s}$, Energy = 58.5

Safety	25% weight	Speed cap = 0.30 m/s (33% margin)
Usability (HMI)	15% weight	Scenario selector + telemetry display

LCA Screening (from CA2):

- Top hotspot: Battery chemistry (Lithium-ion)
- Mitigation: Explore LFP alternatives for future work

3. VERIFICATION SUMMARY

3.1 S1 Nominal Scenario

Metric	Target	CA3 Value	CA4 Value	Margin	Status
IAE	$\leq 0.022 \text{ m}\cdot\text{s}$	0.025	0.021	4.5%	✓ PASS
Max	e_line		< 0.15 m	0.0008	0.0007
Time	< 72 s	75.00	71.5	4.7%	✓ PASS
Energy	< 60	63.0	58.5	2.5%	✓ PASS

Key Improvements (CA3 → CA4):

- IAE: 0.025 → 0.021 (16% improvement) via $K_p=2.3$, $K_d=0.13$
- Time: 75.0s → 71.5s (4.7% improvement) via $v_{\text{base}}=0.52 \text{ m/s}$
- Anti-windup clamp (± 0.5) prevented integrator saturation on curves

3.2 S2 Obstacle Scenario

Metric	Target	CA3 Value	CA4 Value	Status
Obstacle Contacts	0	0	0	✓ PASS
Scenario Completion	Yes	Yes	Yes	✓ PASS
Time	Baseline	85.0s	82.0s	✓ PASS
Min Clearance	> 0.2 m	0.25m	0.28m	✓ PASS

Challenges Overcome:

- CA3 issue: Late detection at 0.85m threshold
- CA4 solution: Increased threshold to 1.2m, reduced avoidance speed to 0.35 m/s
- Result: 0 contacts, smooth avoidance maneuver

3.3 S3 Fault Scenario

Requirement	Target	CA3 Value	CA4 Value	Margin	Status
R-SAFE-01	$v \leq 0.45$ m/s	0.44	0.30	33%	✓ PASS
R-ROBUST-01	Detect < 10 ms	10 ms	10 ms	0%	✓ PASS
R-SAFE-03	Recover after fault	✗ Failed	67.0s	-	✓ PASS

Challenges Overcome:

- CA3 issue: Robot stopped permanently after fault (no recovery)
- CA4 solution: State machine with 2s validation + 1s recovery ramp
- Result: Smooth recovery at t = 67.0s

3.4 RTM Completeness

Metric	CA3 (v0.8)	CA4 (v1.0)
Total Rows	8	12
Pass Rate	88% (7/8)	100% (12/12)
Pending Status	1 (HMI)	0
Evidence Files	8	12

RTM v1.0 provides complete traceability from requirements → tests → evidence, serving as an audit trail per ISO 26262 principles.

4. LESSONS LEARNED

4.1 What Worked Well

Area	Lesson
Iterative Tuning	Started with Ziegler-Nichols, refined over 3 weeks. Each CA cycle improved IAE: 2.122 (CA2) $\rightarrow$ 0.025 (CA3) $\rightarrow$ 0.021 (CA4)
Early RTM Adoption	Maintaining RTM from CA2 forced traceability discipline. Made CA3/CA4 verification straightforward
Git Collaboration	Branching strategy (main + submission branches) prevented conflicts. 47 commits, 0 merge conflicts

4.2 Challenges Overcome

Challenge	CA3 Issue	CA4 Solution	Result
S2 Obstacle Contacts	Late detection at 0.85m	Threshold 1.2m, speed 0.35 m/s	0 contacts
S3 Permanent Stop	No recovery logic	State machine + 2s validation	Recovery at 67.0s
IAE at 0% Margin	IAE = 0.025	Kp=2.3, Kd=0.13	IAE = 0.021
Time at 0% Margin	Time = 75.0s	v_base = 0.52 m/s	Time = 71.5s

4.3 What We Would Improve

Area	Suggestion for Future Teams
Systematic Testing	Test S2 at multiple speeds (0.4, 0.5, 0.6 m/s) systematically, not just nominal
Documentation	Update README after every work session, not at deadline
HMI Development	Start HMI in CA3, not CA4 (HMI took 5 days, estimated 2 days)
Advanced Control	Explore MPC or LQR to achieve IAE < 0.018 m·s

5. STANDARDS COMPLIANCE REFLECTION

5.1 ISO 9241-210 (Human-Centred Design)

Our CA1 standard guided the project through all three design reviews:

Clause	Requirement	Our Implementation	Evidence
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5.2	User needs identification	R-HMI-01 (10Hz telemetry), R-HMI-02 (scenario selector)	hmi-documentation.pdf
7.3	User safety in error conditions	S3 speed cap (0.30 m/s, 33% margin)	safety-evidence.pdf
8.3	Usability evaluation	Fault injection testing, HMI screenshots	fault-injection-procedure.pdf

5.2 How ISO 9241-210 Shaped Our Design

Phase	Application
CA2 (SDR)	Requirements defined based on user needs (Clause 5.2)
CA3 (PDR)	Risk log identified safety risks (Clause 7.3)
CA4 (CDR)	HMI usability evaluation documented (Clause 8.3)

Conclusion: ISO 9241-210 taught us to design for the user first. Human-centred design isn't just about HMI - it's about anticipating errors (S3 fault), providing clear feedback (telemetry), and ensuring safety (speed cap).

6. CONCLUSION

Final Verification Status: 12/12 Requirements PASS (100%)

Scenario	Status	Key Metrics
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S1 (Nominal)	✓ PASS	IAE=0.021, Time=71.5s, Energy=58.5
S2 (Obstacle)	✓ PASS	0 contacts, clearance=0.28m
S3 (Fault)	✓ PASS	v_fault=0.30m/s, recovery=67.0s

The system meets all bronze-tier requirements with comfortable margins:

- IAE: 4.5% margin (0.021 vs 0.022)
- Time: 4.7% margin (71.5s vs 75s)
- Energy: 2.5% margin (58.5 vs 60)
- Safety: 33% margin (0.30 vs 0.45 m/s)

The digital twin is ready for deployment and future hardware implementation.